GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Calificación por traza y tendido (diagnóstica)

Verde: buena. Amarillo: revisión. Rojo: mala. Gris: no evaluable. Política DRAFT; límites provisionales locales.

Calificación diagnóstica condicional del conjunto: MALA

Política TRS_SAMPLE_SCREENING_V1_DRAFT. Cobertura media 75.00 %. La cobertura es disponibilidad de comprobaciones, no confianza estadística. No certifica aceptación de campo.

Conteos diagnósticos: {"BUENA": 2, "REGULAR": 22, "MALA": 0, "NO_EVALUABLE": 0}

Conjunto de registros: NO_EVALUABLE | 24 trazas

Intervalo de buenas: 0-100 %. Fallas demostradas: 0. Evidencia pendiente: 24.

Porcentaje bueno = 100 x buenas / total. BUENA: >=90 %; REGULAR: >=75 % y <90 %; MALA: <75 %. Todas las trazas permanecen en el denominador. El conjunto suma trazas; no promedia porcentajes. Si los extremos posibles cambian de categoría, la evaluación es NO_EVALUABLE. Las abstenciones del picker no califican la adquisición.

Registro	Buenas	Revisión	Malas	No evaluables	Evaluación
KM40_F1_TRS_EXPLORATORY_ASSUMPTIONS	0	21	0	3	NO_EVALUABLE

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Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Registro / tendido: KM40_F1_TRS_EXPLORATORY_ASSUMPTIONS

Calificación diagnóstica del tendido: MALA

Mínimo de cobertura 75 %. Dominios: integridad, morfología, contraste y similitud. Integridad, morfología y contraste son necesarios para declarar buena. Contraste bueno si todo el intervalo alcanza 12 dB; ruido estable si máximo/mínimo <= 4. El intervalo es sensibilidad entre bloques, no un intervalo de confianza.

Mesetas extremas de al menos 3 muestras generan revisión; no son límites ADC estimados. Sin consenso de vecinos, la similitud no es un criterio de falla. Una línea aislada no basta: se comprueba persistencia entre bloques. La evaluación estricta y sus faltantes permanecen separados.

Canal	Diagnóstica	Cobertura	Motivos
ch01	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch02	REGULAR	75 %	Referencia variable entre bloques; contraste sensible al ruido; Ruido de referencia posterior variable o no resuelto: SNR solo condicional
ch03	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch04	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch05	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch06	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch07	BUENA	75 %	Comprobaciones disponibles dentro del escenario diagnóstico declarado
ch08	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch09	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch10	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch11	BUENA	75 %	Comprobaciones disponibles dentro del escenario diagnóstico declarado
ch12	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch13	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch14	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch15	REGULAR	75 %	Contraste intermedio o sensible a los bloques de referencia; Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch16	REGULAR	75 %	Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch17	REGULAR	75 %	Contraste intermedio o sensible a los bloques de referencia; Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica

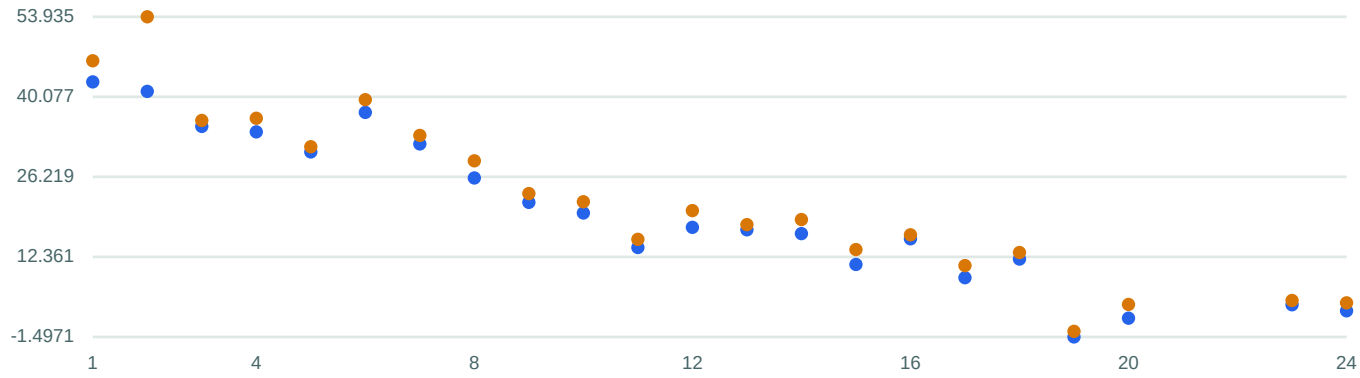
GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

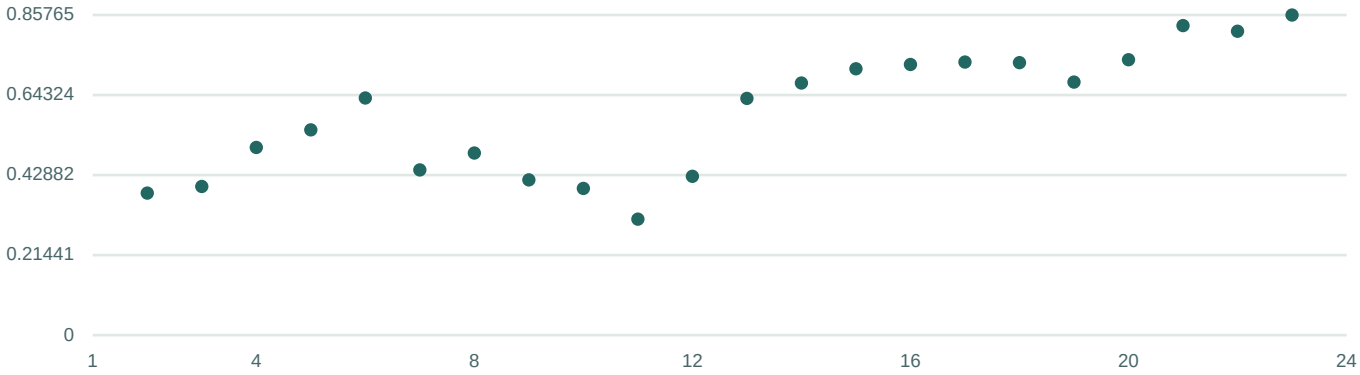
Canal	Diagnóstica	Cobertura	Motivos
ch18	REGULAR	75 %	Contraste intermedio o sensible a los bloques de referencia
ch19	REGULAR	75 %	Contraste condicional bajo; Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch20	REGULAR	75 %	Contraste condicional bajo; Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch21	REGULAR	75 %	Contraste condicional bajo; Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch22	REGULAR	75 %	Contraste condicional bajo; Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch23	REGULAR	75 %	Contraste condicional bajo; Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica
ch24	REGULAR	75 %	Contraste condicional bajo

Contraste estimado: sensibilidad a bloques



Canal | dB condicionales | Inferior azul; Superior naranja

Similitud estimada (sin certificar comparabilidad)



Canal | correlación | |rho|

Criterios de evaluación y umbrales aplicados

Política: TRACE_QUALITY_V1_DRAFT

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Evidencia	Buena	Revisión	Mala / falla
Integridad	Todas las muestras válidas	Huecos parciales	Sin soporte válido
Señal	Variable	Anomalía con modelo declarado	Canal muerto o constante
SNR válida	>=12 dB	>=6 y <12 dB	<6 dB o neta no resuelta con ruido verificado
Clipping	<=0.1 %	>0.1 y <1 %	>=1 %

Se conserva la peor condición demostrada. Solo faltantes: no evaluable. Una falla demostrada prevalece sobre faltantes. SNR requiere independencia y estacionariedad; clipping requiere rails/escala y soporte de apilamiento. La SNR condicional no valida el primer arribo.

Condición	Declaración del protocolo
Ventana de análisis	{"schema_version": "1.4", "start_s": 0.0, "end_s": 2.0}
Ventana de señal	{"schema_version": "1.4", "start_s": 0.0, "end_s": 0.5}
Referencia de ruido	{"schema_version": "1.4", "start_s": 1.5, "end_s": 2.0}

Independencia del ruido: False; estacionariedad: UNKNOWN

Similitud: parámetros y comparabilidad declarados en el protocolo

Parámetro	Valor
arrival	{'schema_version': '1.4', 'start_s': 0.0, 'end_s': 0.5}
baseline	Sin valor
neighbor_radius_m	10.1
min_apparent_velocity_ms	100.0
max_lag_s	0.02
max_neighbors	4
min_neighbors	3
bad_below_rho	0.5
warn_below_rho	0.8
neighbor_consensus_min	0.8
bad_neighbor_fraction_min	0.75
baseline_ratio_warn	5.0
comparable_channels	False

Ruido: protocolo y límites de soporte

Parámetro	Valor

GeoQC - TRS | Calidad por traza y tendido

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Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Parámetro	Valor
low_band_hz	[1.0, 15.0]
minimum_cycles	3.0
noise_blocks	4
block_power_ratio_warn	4.0
low_ratio_warn	0.5
mains_candidates_hz	[50.0, 60.0]
harmonics	5
line_prominence_db	10.0
reference_kind	POST_SIGNAL

GeoQC - TRS | Calidad por traza y tendido

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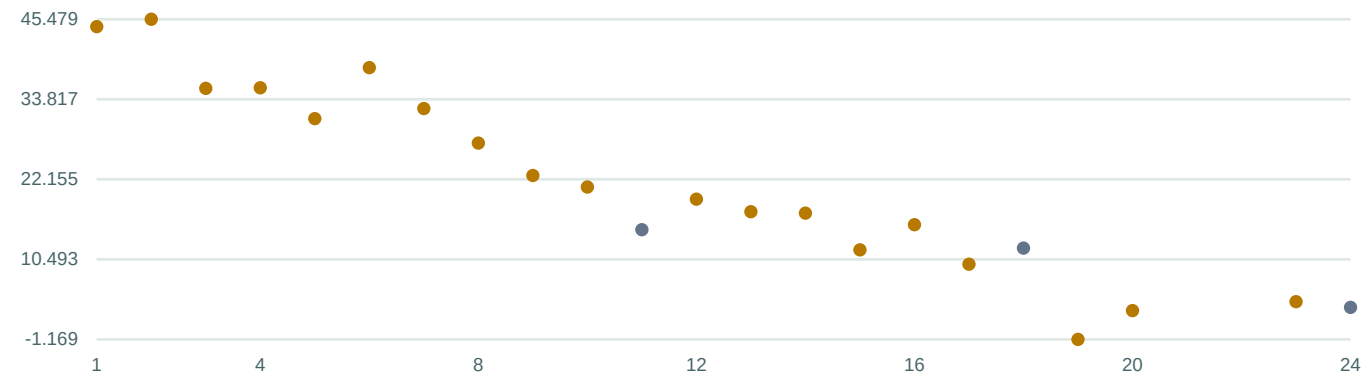
Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Diagnóstico gráfico de calidad y ruido

SNR neta válida por canal

Sin valores evaluables con este protocolo; no se representan como cero.

SNR condicional de ventana



Canal | dB | No valida el primer arribo

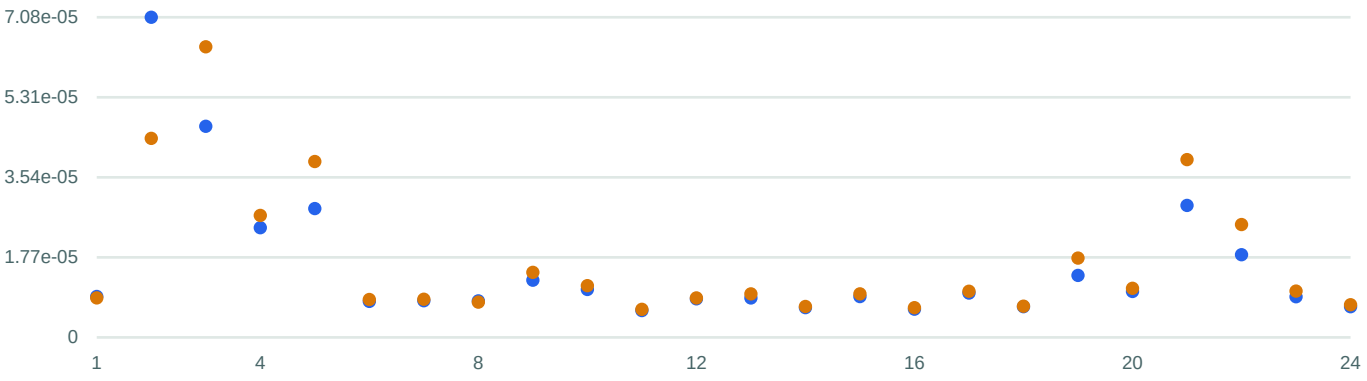
Clipping confirmado

Sin valores evaluables con este protocolo; no se representan como cero.

Similitud local y consenso

Sin valores evaluables con este protocolo; no se representan como cero.

Escalas de ruido: RMS AC y MAD



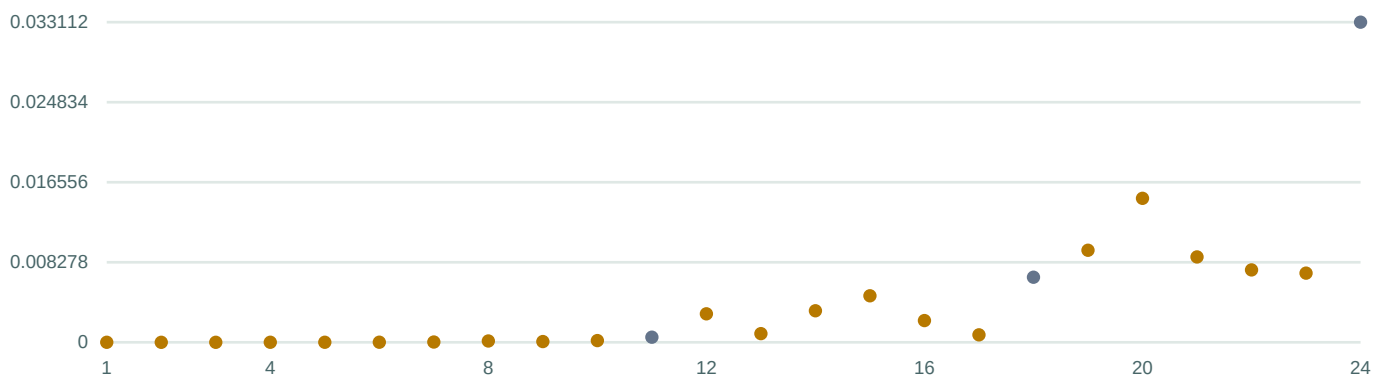
Canal | V | RMS AC azul; MAD naranja

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

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Ruido LF / potencia total de señal



Canal | razón de potencias | Banda soportada parcial

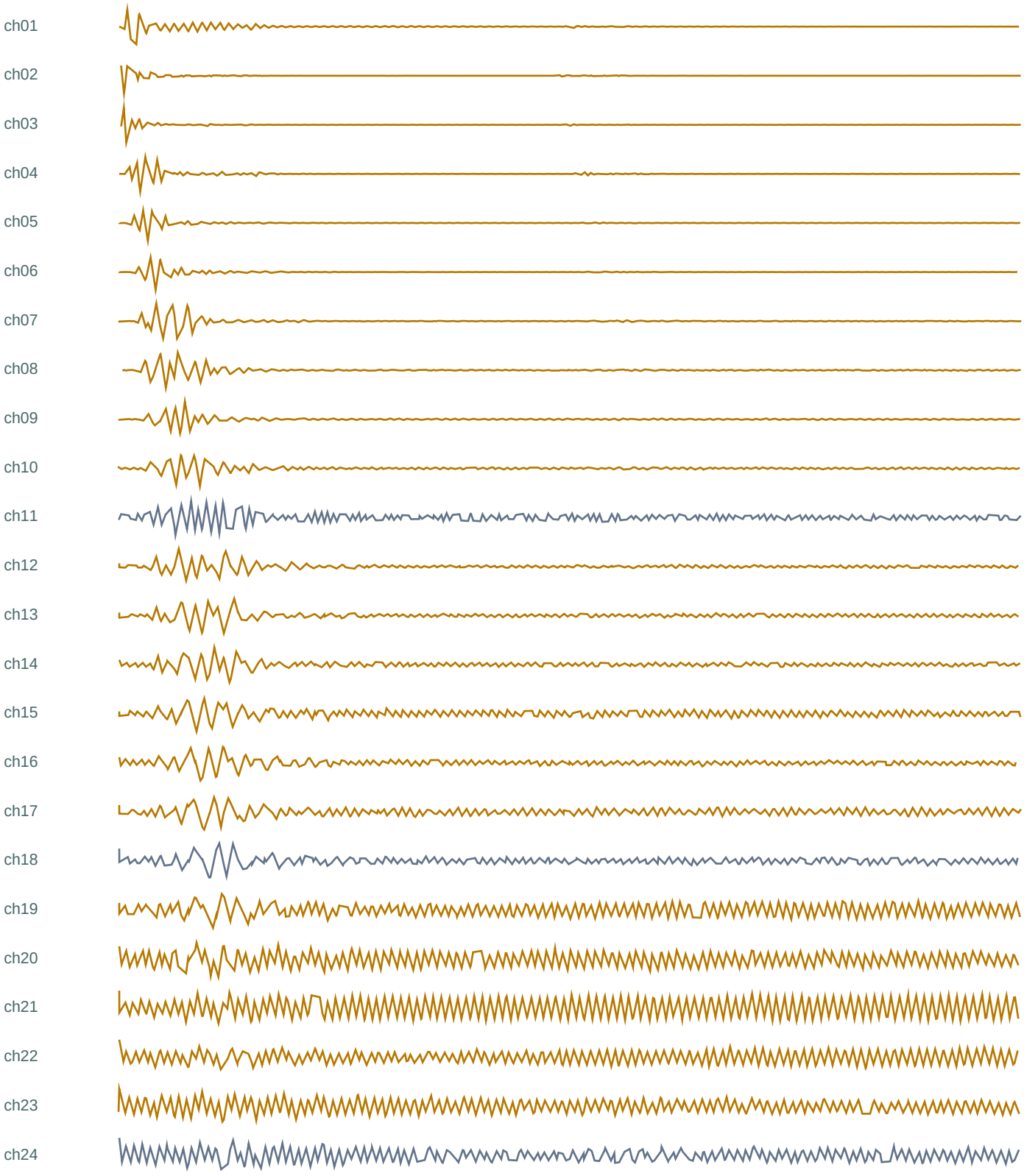
GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

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Formas de onda por canal

Envolvente min/max de las muestras originales. Centrado y normalización por canal solo para visualizar; sin filtrado, alineación ni corrección de polaridad.



Tiempo registrado: 0 a 2 s (primer canal).

GeoQC - TRS | Calidad por traza y tendido

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Diagnóstico por canal: métricas, motivos y acciones

ch01: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [42.661459479338866, 46.318977433139416], "net_power_resolved": [true, true], "reference_power_ratio": 2.321338110487312, "reference_stable": true, "normalized_block_powers": [1.0479253833433675e-06, 4.514316025782989e-07, 5.389713221539677e-07, 7.389650175259832e-07], "normalization_amplitude": 0.010805375874042511, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 60.0, "block_prominence_db": [13.140110228362138, 15.775642297624323, -3.4510328502435748, 15.62612043922855], "fraction": 0.75, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}],

"estimated_similarity": {"state": "MISSING", "grade": "REGULAR", "median_abs_rho": null, "neighbor_consensus_rho": null, "pairs": [{"abs_rho": 0.3806029028020091, "signed_rho": 0.3806029028020091, "lag_samples": 125, "overlap_samples": 3875, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.015625, "neighbor_channel_1based": 2, "distance_m": 5.0}, {"abs_rho": 0.3925975587599119, "signed_rho": 0.3925975587599119, "lag_samples": 72, "overlap_samples": 3928, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.009000000000000001, "neighbor_channel_1based": 3, "distance_m": 10.0}], "reasons": ["Vecinos comparables insuficientes; sin decisión local"], "spectral_median_hz": 50.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 48.59335888528038, "spectral_bandwidth_hz": 12.89691160619992, "arrival_rms_ac": 0.0015027605928369574, "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": null, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 44.403 dB. RMS AC: 9.0517e-06; MAD: 8.7348e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [1.2235173091935898e-10, 5.270741490289582e-11, 6.292821533823452e-11, 8.62787087900257e-11], "max_min_power_ratio": 2.321338110487312, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 3.553583354293753e-06, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 58.0, "prominence_db": 11.162197968960122, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

GeoQC - TRS | Calidad por traza y tendido

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ch02: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Ruido de referencia posterior variable o no resuelto: SNR solo condicional; Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Referencia variable entre bloques; contraste sensible al ruido", "Ruido de referencia posterior variable o no resuelto: SNR solo condicional"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [41.02056665356538, 53.93474105080923], "net_power_resolved": [true, true], "reference_power_ratio": 19.560721148433732, "reference_stable": false, "normalized_block_powers": [5.244704971328031e-08, 1.5392386362526831e-07, 2.341416266205232e-07, 1.0259021144995174e-06], "normalization_amplitude": 0.11679104715585709, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 150.0, "block_prominence_db": [9.97910755654064, -3.2611148134154693, 9.538455517292272, 7.528453576990603], "fraction": 0.0, "state": "ESTIMATED", "persistent_candidate":

false, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.3806029028020091, "neighbor_consensus_rho": 0.3925975587599119, "pairs": [{"abs_rho": 0.3806029028020091, "signed_rho": 0.3806029028020091, "lag_samples": -125, "overlap_samples": 3875, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.015625, "neighbor_channel_1based": 1, "distance_m": 5.0}, {"abs_rho": 0.682129327135893, "signed_rho": 0.682129327135893, "lag_samples": -46, "overlap_samples": 3954, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.00575, "neighbor_channel_1based": 3, "distance_m": 5.0}, {"abs_rho": 0.16863699168939603, "signed_rho": 0.16863699168939603, "lag_samples": -160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": -0.02, "neighbor_channel_1based": 4, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 94.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 81.7817443149372, "spectral_bandwidth_hz": 48.59181844092074, "arrival_rms_ac": 0.013304789688716621, "low_correlation_fraction": 0.6666666666666666, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en

campo"], "local_amplitude_ratio": 8.85356573238285, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 45.479 dB. RMS AC: 7.08e-05; MAD: 4.4017e-05 V.

Estacionariedad interna: {"state": "VARIABLE_OR_UNRESOLVED", "block_powers": [7.153855567431453e-10, 2.099544387674795e-09, 3.193726602971436e-09, 1.3993457389069695e-08], "max_min_power_ratio": 19.56072114843375, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 5.517746798532021e-08, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 150.0, "fundamental_hz": 50.0, "harmonic": 3, "estimated_hz": 152.1974465227861, "prominence_db": 10.6122807390684, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Revisar subventanas de ruido y repetir una toma de ruido antes del disparo; Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch03: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [34.956300988304776, 35.98418428436297], "net_power_resolved": [true, true], "reference_power_ratio": 1.2669489063525003, "reference_stable": true, "normalized_block_powers": [4.65563135179913e-06, 4.273594965025799e-06, 4.144272757167489e-06, 3.6746796405567154e-06], "normalization_amplitude": 0.02279726229608059, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [30.840767835121753, 30.668167350738965, 25.272140476399905, 26.271900246826604], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}],

"estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.3983365854716101, "neighbor_consensus_rho": 0.3263695802302914, "pairs": [{"abs_rho": 0.682129327135893, "signed_rho": 0.682129327135893, "lag_samples": 46, "overlap_samples": 3954, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.00575, "neighbor_channel_1based": 2, "distance_m": 5.0}, {"abs_rho": 0.4040756121833083, "signed_rho": 0.4040756121833083, "lag_samples": -103, "overlap_samples": 3897, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.012875000000000001, "neighbor_channel_1based": 4, "distance_m": 5.0}, {"abs_rho": 0.3925975587599119, "signed_rho": 0.3925975587599119, "lag_samples": -72, "overlap_samples": 3928, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.009000000000000001, "neighbor_channel_1based": 1, "distance_m": 10.0}, {"abs_rho": 0.3719944152057629, "signed_rho": -0.3719944152057629, "lag_samples": -142, "overlap_samples": 3858, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.017750000000000002, "neighbor_channel_1based": 5, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 44.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 47.538873665664596, "spectral_bandwidth_hz":

30.746212879839934, "arrival_rms_ac": 0.002752684192498654, "low_correlation_fraction": 0.75, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 1.8680356315731546, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 35.409 dB. RMS AC: 4.669e-05; MAD: 6.4268e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [2.419602231060242e-09, 2.2210521260512342e-09, 2.153841413042636e-09, 1.9097867474594435e-09], "max_min_power_ratio": 1.266948906352501, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 7.281536867664233e-07, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 50.49850247628047, "prominence_db": 30.131687749683923, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch04: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [34.00641303602174, 36.36484565031623], "net_power_resolved": [true, true], "reference_power_ratio": 1.7209606619406939, "reference_stable": true, "normalized_block_powers": [6.856283625149035e-06, 7.648738583542538e-06, 6.755100378982101e-06, 1.1625262019688869e-05], "normalization_amplitude": 0.008444499224424362, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [21.789929582391103, 15.82364165762997, 12.190421281416517, 19.074154892707252], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}, {"nominal_hz":

120.0, "block_prominence_db": [2.708596638150106, 8.559642443652828, 1.5388198691104353, 13.181339679540717], "fraction": 0.25, "state": "ESTIMATED", "persistent_candidate": false, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.5028798981420182, "neighbor_consensus_rho": 0.2426909054326358, "pairs": [{"abs_rho": 0.4040756121833083, "signed_rho": 0.4040756121833083, "lag_samples": 103, "overlap_samples": 3897, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.012875000000000001, "neighbor_channel_1based": 3, "distance_m": 5.0}, {"abs_rho": 0.7197537991073464, "signed_rho": 0.7197537991073464, "lag_samples": -129, "overlap_samples": 3871, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.016125, "neighbor_channel_1based": 5, "distance_m": 5.0}, {"abs_rho": 0.16863699168939603, "signed_rho": 0.16863699168939603, "lag_samples": 160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": 0.02, "neighbor_channel_1based": 2, "distance_m": 10.0}, {"abs_rho": 0.601684184100728, "signed_rho": -0.601684184100728, "lag_samples": 28, "overlap_samples": 3972, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.0035, "neighbor_channel_1based": 6, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 50.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of

first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 53.178492116770315, "spectral_bandwidth_hz": 21.48078355114011, "arrival_rms_ac": 0.0014443825408092658, "low_correlation_fraction": 0.5, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.7675133390846715, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 35.495 dB. RMS AC: 2.4259e-05; MAD: 2.6975e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [4.889186175759487e-10, 5.454282376458938e-10, 4.817032840888208e-10, 8.289924026445033e-10], "max_min_power_ratio": 1.7209606619406943, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 4.991138643323409e-06, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 47.42126893901361, "prominence_db": 17.572032907285514, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 120.0, "fundamental_hz": 60.0, "harmonic": 2, "estimated_hz": 117.0, "prominence_db": 10.250958424308473, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch05: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [30.534975987872322, 31.435863187673846], "net_power_resolved": [true, true], "reference_power_ratio": 1.2303164990241187, "reference_stable": true, "normalized_block_powers": [1.2767343300225671e-05, 1.3875049608234516e-05, 1.170826128640426e-05, 1.127762621995246e-05], "normalization_amplitude": 0.008067556656897068, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [25.428197645953148, 28.075858752063752, 27.95998573264854, 22.464376500460958], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}],

"estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.5499979971485303, "neighbor_consensus_rho": 0.31346143746806027, "pairs": [{"abs_rho": 0.7197537991073464, "signed_rho": 0.7197537991073464, "lag_samples": 129, "overlap_samples": 3871, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.016125, "neighbor_channel_1based": 4, "distance_m": 5.0}, {"abs_rho": 0.7769204004253462, "signed_rho": 0.7769204004253462, "lag_samples": -136, "overlap_samples": 3864, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.017, "neighbor_channel_1based": 6, "distance_m": 5.0}, {"abs_rho": 0.3719944152057629, "signed_rho": -0.3719944152057629, "lag_samples": 142, "overlap_samples": 3858, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.017750000000000002, "neighbor_channel_1based": 3, "distance_m": 10.0}, {"abs_rho": 0.380242195189714, "signed_rho": -0.380242195189714, "lag_samples": -160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": -0.02, "neighbor_channel_1based": 7, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 56.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 54.0072912955791, "spectral_bandwidth_hz": 25.563293261408326, "arrival_rms_ac":

0.0010111136915402644, "low_correlation_fraction": 0.5, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.9587685116708714, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 30.998 dB. RMS AC: 2.8491e-05; MAD: 3.8897e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [8.309685446098021e-10, 9.030641307451682e-10, 7.620376935350888e-10, 7.34009607659065e-10], "max_min_power_ratio": 1.2303164990241193, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 4.972914075588652e-06, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 49.818936406547785, "prominence_db": 35.08441932863026, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch06: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [37.37716301279568, 39.5847360037569], "net_power_resolved": [true, true], "reference_power_ratio": 1.6623621638330037, "reference_stable": true, "normalized_block_powers": [4.0038711046627865e-06, 6.4486788352911305e-06, 5.8114664236164935e-06, 3.879226185238746e-06], "normalization_amplitude": 0.0035404975060373545, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [10.299972867827712, 12.720199798138975, 18.308777099975554, 13.068097598238527], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}],

"estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.6353989030156528, "neighbor_consensus_rho": 0.3502371094093427, "pairs": [{"abs_rho": 0.7769204004253462, "signed_rho": 0.7769204004253462, "lag_samples": 136, "overlap_samples": 3864, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.017, "neighbor_channel_1based": 5, "distance_m": 5.0}, {"abs_rho": 0.6691136219305776, "signed_rho": 0.6691136219305776, "lag_samples": -119, "overlap_samples": 3881, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.014875000000000001, "neighbor_channel_1based": 7, "distance_m": 5.0}, {"abs_rho": 0.601684184100728, "signed_rho": -0.601684184100728, "lag_samples": -28, "overlap_samples": 3972, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.0035, "neighbor_channel_1based": 4, "distance_m": 10.0}, {"abs_rho": 0.47071840910223695, "signed_rho": 0.47071840910223695, "lag_samples": 81, "overlap_samples": 3919, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.010125, "neighbor_channel_1based": 8, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 44.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 46.17689095244257, "spectral_bandwidth_hz": 27.93070141186585, "arrival_rms_ac":

0.0006648099894672441, "low_correlation_fraction": 0.25, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.9833732698155344, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 38.429 dB. RMS AC: 7.9652e-06; MAD: 8.3828e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [5.0189015132534655e-11, 8.083497974556829e-11, 7.284744404919359e-11, 4.862657578730163e-11], "max_min_power_ratio": 1.6623621638330037, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 1.0031494956636824e-05, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 53.0, "prominence_db": 16.994389275380644, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch07: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Estimaciones y soporte diagnóstico: {"grade": "BUENA", "color": "#16803c", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Comprobaciones disponibles dentro del escenario diagnóstico declarado"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [31.920173831191548, 33.3943221949156], "net_power_resolved": [true, true], "reference_power_ratio": 1.4038947390363123, "reference_stable": true, "normalized_block_powers": [1.4341873160629946e-06, 1.6308318350569838e-06, 1.4072858882285067e-06, 1.1616482273994772e-06], "normalization_amplitude": 0.006766832433640957, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [8.19417282098421, 13.1587754126668, 11.8909130738356, 5.415421376827872], "fraction": 0.5, "state": "ESTIMATED", "persistent_candidate": false, "cause": "UNIDENTIFIED"}, {"nominal_hz": 60.0, "block_prominence_db":

[5.781903706668531, 13.920106958612715, 9.371454751121837, 4.653745647287414], "fraction": 0.25, "state": "ESTIMATED", "persistent_candidate": false, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.4426851879362664, "neighbor_consensus_rho": 0.3954752163656042, "pairs": [{"abs_rho": 0.6691136219305776, "signed_rho": 0.6691136219305776, "lag_samples": 119, "overlap_samples": 3881, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.014875000000000001, "neighbor_channel_1based": 6, "distance_m": 5.0}, {"abs_rho": 0.5051281806828187, "signed_rho": 0.5051281806828187, "lag_samples": -83, "overlap_samples": 3917, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.010375, "neighbor_channel_1based": 8, "distance_m": 5.0}, {"abs_rho": 0.380242195189714, "signed_rho": -0.380242195189714, "lag_samples": 160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": 0.02, "neighbor_channel_1based": 5, "distance_m": 10.0}, {"abs_rho": 0.24293454928476071, "signed_rho": -0.24293454928476071, "lag_samples": -16, "overlap_samples": 3984, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.002, "neighbor_channel_1based": 9, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 32.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy.

Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 37.83658601144548, "spectral_bandwidth_hz": 29.987264954984802, "arrival_rms_ac": 0.0003409873058601473, "low_correlation_fraction": 0.5, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.7975666637397475, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 32.471 dB. RMS AC: 8.1108e-06; MAD: 8.438e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [6.567146758574747e-11, 7.467582427639146e-11, 6.44396506353001e-11, 5.31918969421110 34e-11], "max_min_power_ratio": 1.4038947390363135, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 3.0017455559916238e-05, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 51.805710016396596, "prominence_db": 13.657805867991064, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 10.909260593088018, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch08: REGULAR

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [26.02827968834145, 28.996427989947925], "net_power_resolved": [true, true], "reference_power_ratio": 1.9782410617423827, "reference_stable": true, "normalized_block_powers": [1.4434810833165327e-05, 1.9796113346131914e-05, 1.88479369514735e-05, 2.8555535508651423e-05], "normalization_amplitude": 0.0017764177173376083, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [8.975932342007466, 15.455007687875746, 17.632366506679634, 16.784847131489773], "fraction": 0.75, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}, {"nominal_hz":

60.0, "block_prominence_db": [7.336042870720768, 12.156490125832596, 17.632366506679634, 11.674114092078947], "fraction": 0.75, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.48792329489252784, "neighbor_consensus_rho": 0.2853089627296619, "pairs": [{"abs_rho": 0.5051281806828187, "signed_rho": 0.5051281806828187, "lag_samples": 83, "overlap_samples": 3917, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.010375, "neighbor_channel_1based": 7, "distance_m": 5.0}, {"abs_rho": 0.6494926717963164, "signed_rho": 0.6494926717963164, "lag_samples": -107, "overlap_samples": 3893, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.013375, "neighbor_channel_1based": 9, "distance_m": 5.0}, {"abs_rho": 0.47071840910223695, "signed_rho": 0.47071840910223695, "lag_samples": -81, "overlap_samples": 3919, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.010125, "neighbor_channel_1based": 6, "distance_m": 10.0}, {"abs_rho": 0.24910632590494924, "signed_rho": -0.24910632590494924, "lag_samples": 116, "overlap_samples": 3884, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.0145, "neighbor_channel_1based": 10, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 36.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of

first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 40.08177859517325, "spectral_bandwidth_hz": 44.31900308735847, "arrival_rms_ac": 0.00019025911349971285, "low_correlation_fraction": 0.5, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.7397751456021099, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 27.435 dB. RMS AC: 8.0762e-06; MAD: 7.7993e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [4.555135380371245e-11, 6.246980119034307e-11, 5.947767895745732e-11, 9.011155851245908e-11], "max_min_power_ratio": 1.9782410617423833, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.00013714966238515078, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 51.69513302678502, "prominence_db": 15.143867806297937, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 15.895302668251015, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch09: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [21.80234388782644, 23.32607827994704], "net_power_resolved": [true, true], "reference_power_ratio": 1.4175212124258678, "reference_stable": true, "normalized_block_powers": [0.00019065963323799495, 0.00015346593754626402, 0.0001373467396526349, 0.00013450213765176 08], "normalization_amplitude": 0.0010170218301936984, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [24.519492726631135, 29.644171795972852, 21.71353386322965, 25.296288890109608], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}, {"nominal_hz":

60.0, "block_prominence_db": [24.326460713161076, 27.70071713622841, 22.168530647391197, 21.235439734019575], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.4160224760332003, "neighbor_consensus_rho": 0.3279465085965483, "pairs": [{"abs_rho": 0.6494926717963164, "signed_rho": 0.6494926717963164, "lag_samples": 107, "overlap_samples": 3893, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.013375, "neighbor_channel_1based": 8, "distance_m": 5.0}, {"abs_rho": 0.3931457602501098, "signed_rho": 0.3931457602501098, "lag_samples": -115, "overlap_samples": 3885, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.014375, "neighbor_channel_1based": 10, "distance_m": 5.0}, {"abs_rho": 0.24293454928476071, "signed_rho": -0.24293454928476071, "lag_samples": 16, "overlap_samples": 3984, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.002, "neighbor_channel_1based": 7, "distance_m": 10.0}, {"abs_rho": 0.4388991918162908, "signed_rho": 0.4388991918162908, "lag_samples": -96, "overlap_samples": 3904, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.012, "neighbor_channel_1based": 11, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 46.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-

arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 43.432745237620225, "spectral_bandwidth_hz": 40.0379349079706, "arrival_rms_ac": 0.00017338280952095055, "low_correlation_fraction": 0.75, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 1.1180169314365858, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 22.712 dB. RMS AC: 1.2654e-05; MAD: 1.4384e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [1.9720562727904959e-10, 1.5873494544070754e-10, 1.4206232062830028e-10, 1.39120053760298e-10], "max_min_power_ratio": 1.4175212124258683, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 7.771845589739662e-05, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 48.069414822215954, "prominence_db": 30.74826585116025, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 18.815742783093953, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch10: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [19.963125895403625, 21.91060921213378], "net_power_resolved": [true, true], "reference_power_ratio": 1.5601937167488542, "reference_stable": true, "normalized_block_powers": [1.776517679851979e-06, 2.112940334167039e-06, 2.6422909552836823e-06, 1.6935659507652116e-06], "normalization_amplitude": 0.007370578590780497, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 2, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [16.986047703737146, 15.949721957728158, 17.346723760559595, 14.964233512607066], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}, {"nominal_hz":

60.0, "block_prominence_db": [16.986047703737146, 19.592563461407355, 20.411897560896325, 14.964233512607066], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.39311604633993114, "neighbor_consensus_rho": 0.38355441641741217, "pairs": [{"abs_rho": 0.3931457602501098, "signed_rho": 0.3931457602501098, "lag_samples": 115, "overlap_samples": 3885, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.014375, "neighbor_channel_1based": 9, "distance_m": 5.0}, {"abs_rho": 0.3930863324297525, "signed_rho": -0.3930863324297525, "lag_samples": -100, "overlap_samples": 3900, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.0125, "neighbor_channel_1based": 11, "distance_m": 5.0}, {"abs_rho": 0.24910632590494924, "signed_rho": -0.24910632590494924, "lag_samples": -116, "overlap_samples": 3884, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.0145, "neighbor_channel_1based": 8, "distance_m": 10.0}, {"abs_rho": 0.45064760939485954, "signed_rho": -0.45064760939485954, "lag_samples": 135, "overlap_samples": 3865, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.016875, "neighbor_channel_1based": 12, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 38.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration

of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 39.023342836571366, "spectral_bandwidth_hz": 56.50615745170187, "arrival_rms_ac": 0.00011990221705932234, "low_correlation_fraction": 1.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.9508272651978331, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 21.034 dB. RMS AC: 1.0603e-05; MAD: 1.1443e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [9.651008466278108e-11, 1.1478638960398988e-10, 1.4354358906204423e-10, 9.20036964135206e-11], "max_min_power_ratio": 1.5601937167488575, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.00017853718499126156, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 49.29787713375482, "prominence_db": 13.6573618071094, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 14.980731851998419, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch11: NO_EVALUABLE

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "BUENA", "color": "#16803c", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Comprobaciones disponibles dentro del escenario diagnóstico declarado"], "strict_grade": "NO_EVALUABLE", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [13.982726857234178, 15.40073899512813], "net_power_resolved": [true, true], "reference_power_ratio": 1.3712813434599813, "reference_stable": true, "normalized_block_powers": [0.00032161288447816186, 0.00041125414007611033, 0.00031554409459660845, 0.000299905006392375], "normalization_amplitude": 0.00032094071502797306, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.310796187434781, "neighbor_consensus_rho": 0.3552297348851881, "pairs": [{"abs_rho": 0.3930863324297525, "signed_rho": -0.3930863324297525, "lag_samples": 100,

"overlap_samples": 3900, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.0125, "neighbor_channel_1based": 10, "distance_m": 5.0}, {"abs_rho": 0.22850604243980946, "signed_rho": -0.22850604243980946, "lag_samples": 159, "overlap_samples": 3841, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.019875, "neighbor_channel_1based": 12, "distance_m": 5.0}, {"abs_rho": 0.4388991918162908, "signed_rho": 0.4388991918162908, "lag_samples": 96, "overlap_samples": 3904, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.012, "neighbor_channel_1based": 9, "distance_m": 10.0}, {"abs_rho": 0.1137785631287736, "signed_rho": -0.1137785631287736, "lag_samples": 61, "overlap_samples": 3939, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.007625, "neighbor_channel_1based": 13, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 58.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 69.5083889102028, "spectral_bandwidth_hz": 160.41714445525037, "arrival_rms_ac": 3.319912018734117e-05, "low_correlation_fraction": 1.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.3341203752550681,

"quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 14.812 dB. RMS AC: 5.9357e-06; MAD: 6.2078e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [3.312707346731763e-11, 4.236038656891842e-11, 3.2501970251723084e-11, 3.089109814768994e-11], "max_min_power_ratio": 1.3712813434599818, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.0005297703907517631, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: []

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch12: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [17.478443388709312, 20.375841918755683], "net_power_resolved": [true, true], "reference_power_ratio": 1.932020571264765, "reference_stable": true, "normalized_block_powers": [3.11933404247658e-06, 5.320719942655062e-06, 2.7636241064553273e-06, 2.753966506253058e-06], "normalization_amplitude": 0.004527940880507231, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [9.978362578694327, 18.767849930075784, 14.331015853018474, 15.911263138781402], "fraction": 0.75, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}, {"nominal_hz":

60.0, "block_prominence_db": [13.590755011946127, 19.9919538075566, 9.932523754518805, 15.506299063973989], "fraction": 0.75, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.4255754614443571, "neighbor_consensus_rho": 0.3105818301467974, "pairs": [{"abs_rho": 0.22850604243980946, "signed_rho": -0.22850604243980946, "lag_samples": -159, "overlap_samples": 3841, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.019875, "neighbor_channel_1based": 11, "distance_m": 5.0}, {"abs_rho": 0.5729745681461759, "signed_rho": 0.5729745681461759, "lag_samples": -92, "overlap_samples": 3908, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.0115, "neighbor_channel_1based": 13, "distance_m": 5.0}, {"abs_rho": 0.45064760939485954, "signed_rho": -0.45064760939485954, "lag_samples": -135, "overlap_samples": 3865, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.016875, "neighbor_channel_1based": 10, "distance_m": 10.0}, {"abs_rho": 0.40050331349385465, "signed_rho": 0.40050331349385465, "lag_samples": -160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": -0.02, "neighbor_channel_1based": 14, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 30.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration

of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 34.80810001030977, "spectral_bandwidth_hz": 80.25303423313572, "arrival_rms_ac": 7.882328815416187e-05, "low_correlation_fraction": 0.75, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 1.3854238935984473, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 19.264 dB. RMS AC: 8.5291e-06; MAD: 8.7307e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [6.39533620594762e-11, 1.0908672308770538e-10, 5.666050851550037e-11, 5.6462505995106235e-11], "max_min_power_ratio": 1.9320205712647651, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.0029469789028540045, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 53.0, "prominence_db": 19.116393252200194, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 16.96951810326547, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch13: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [17.05264168763354, 17.945051567179437], "net_power_resolved": [true, true], "reference_power_ratio": 1.2237106832025322, "reference_stable": true, "normalized_block_powers": [3.577683167469181e-05, 3.351453132988403e-05, 2.923634823638335e-05, 3.0623646468088765e-05], "normalization_amplitude": 0.0015226961113512516, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 60.0, "block_prominence_db": [16.240675833838246, 15.707177593680841, 19.400239654564825, 17.310385911343083], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}],

"estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.6342949098078281, "neighbor_consensus_rho": 0.20684195713400005, "pairs": [{"abs_rho": 0.5729745681461759, "signed_rho": 0.5729745681461759, "lag_samples": 92, "overlap_samples": 3908, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.0115, "neighbor_channel_1based": 12, "distance_m": 5.0}, {"abs_rho": 0.7659808388249166, "signed_rho": 0.7659808388249166, "lag_samples": -86, "overlap_samples": 3914, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.010750000000000001, "neighbor_channel_1based": 14, "distance_m": 5.0}, {"abs_rho": 0.1137785631287736, "signed_rho": -0.1137785631287736, "lag_samples": -61, "overlap_samples": 3939, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.007625, "neighbor_channel_1based": 11, "distance_m": 10.0}, {"abs_rho": 0.6956152514694803, "signed_rho": 0.6956152514694803, "lag_samples": -147, "overlap_samples": 3853, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.018375, "neighbor_channel_1based": 15, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 24.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 29.33075880567677, "spectral_bandwidth_hz": 75.9260637552332, "arrival_rms_ac":

6.550660269257185e-05, "low_correlation_fraction": 0.25, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 1.5438029148746293, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 17.437 dB. RMS AC: 8.721e-06; MAD: 9.6239e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [8.295228526243435e-11, 7.770690788362774e-11, 6.778749781389723e-11, 7.100409227667366e-11], "max_min_power_ratio": 1.2237106832025324, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.000897339418323071, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 20.425611759130913, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch14: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [16.39133639704033, 18.8273444686719], "net_power_resolved": [true, true], "reference_power_ratio": 1.7353886961665104, "reference_stable": true, "normalized_block_powers": [1.1269983431450076e-05, 1.3312908583944246e-05, 1.065091883969155e-05, 7.671427509786472e-06], "normalization_amplitude": 0.00198226235806942, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 60.0, "block_prominence_db": [14.243680113621803, 12.949201432341738, 12.29151694502604, 17.68687967606037], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}],

"estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.6754995461878568, "neighbor_consensus_rho": 0.45527073515034167, "pairs": [{"abs_rho": 0.7659808388249166, "signed_rho": 0.7659808388249166, "lag_samples": 86, "overlap_samples": 3914, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.010750000000000001, "neighbor_channel_1based": 13, "distance_m": 5.0}, {"abs_rho": 0.7316191745048959, "signed_rho": 0.7316191745048959, "lag_samples": -62, "overlap_samples": 3938, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.00775, "neighbor_channel_1based": 15, "distance_m": 5.0}, {"abs_rho": 0.40050331349385465, "signed_rho": 0.40050331349385465, "lag_samples": 160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": 0.02, "neighbor_channel_1based": 12, "distance_m": 10.0}, {"abs_rho": 0.6193799178708178, "signed_rho": 0.6193799178708178, "lag_samples": -143, "overlap_samples": 3857, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.017875000000000002, "neighbor_channel_1based": 16, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 30.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 34.983406804493676, "spectral_bandwidth_hz": 102.20388650522278,

"arrival_rms_ac": 4.82828136995899e-05, "low_correlation_fraction": 0.25, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.9338737345404188, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 17.226 dB. RMS AC: 6.583e-06; MAD: 6.839e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [4.4283867809722944e-11, 5.2311264473479124e-11, 4.185133765438911e-11, 3.0143831516844135e-11], "max_min_power_ratio": 1.735388696166511, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.0032626544528509523, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 16.066967760564275, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

ch15: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste intermedio o sensible a los bloques de referencia", "Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [11.0510836582666, 13.629296443584561], "net_power_resolved": [true, true], "reference_power_ratio": 1.7515918915354889, "reference_stable": true, "normalized_block_powers": [0.0006647651006811732, 0.0006438331297816191, 0.0005236372256725993, 0.0003795205400833545], "normalization_amplitude": 0.0003827876935247332, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 60.0, "block_prominence_db": [13.313603584108408, 11.372883852530995, 11.666634063537892, 14.51701449296041], "fraction": 1.0, "state": "ESTIMATED",

"persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.713617212987188, "neighbor_consensus_rho": 0.47847341001266264, "pairs": [{"abs_rho": 0.7316191745048959, "signed_rho": 0.7316191745048959, "lag_samples": 62, "overlap_samples": 3938, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.00775, "neighbor_channel_1based": 14, "distance_m": 5.0}, {"abs_rho": 0.8539135525541941, "signed_rho": 0.8539135525541941, "lag_samples": -70, "overlap_samples": 3930, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.00875, "neighbor_channel_1based": 16, "distance_m": 5.0}, {"abs_rho": 0.6956152514694803, "signed_rho": 0.6956152514694803, "lag_samples": 147, "overlap_samples": 3853, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.018375, "neighbor_channel_1based": 13, "distance_m": 10.0}, {"abs_rho": 0.5909644572500682, "signed_rho": 0.5909644572500682, "lag_samples": -141, "overlap_samples": 3859, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.017625000000000002, "neighbor_channel_1based": 17, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 26.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 35.75682488516573,

"spectral_bandwidth_hz": 140.62532666649741, "arrival_rms_ac": 3.658113112317472e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.848951915138832, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 11.878 dB. RMS AC: 9.0301e-06; MAD: 9.6295e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [9.740564922294798e-11, 9.433856249878375e-11, 7.672668717367789e-11, 5.560978541500314e-11], "max_min_power_ratio": 1.7515918915354887, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.004811487007203605, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 22.57899516116083, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch16: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [15.495477674795232, 16.18558710282031], "net_power_resolved": [true, true], "reference_power_ratio": 1.167499213846391, "reference_stable": true, "normalized_block_powers": [1.5871065145917443e-05, 1.4849024840031942e-05, 1.5258694375603794e-05, 1.3594069235926366e-05], "normalization_amplitude": 0.0015757338842377067, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 60.0, "block_prominence_db": [12.196606300813063, 13.264209021390593, 19.73087958896003, 18.732569631857807], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}],

"estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.7251571252744449, "neighbor_consensus_rho": 0.41064797236511785, "pairs": [{"abs_rho": 0.8539135525541941, "signed_rho": 0.8539135525541941, "lag_samples": 70, "overlap_samples": 3930, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.00875, "neighbor_channel_1based": 15, "distance_m": 5.0}, {"abs_rho": 0.7868267374044005, "signed_rho": 0.7868267374044005, "lag_samples": -68, "overlap_samples": 3932, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.0085, "neighbor_channel_1based": 17, "distance_m": 5.0}, {"abs_rho": 0.6193799178708178, "signed_rho": 0.6193799178708178, "lag_samples": 143, "overlap_samples": 3857, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.017875000000000002, "neighbor_channel_1based": 14, "distance_m": 10.0}, {"abs_rho": 0.6634875131444893, "signed_rho": 0.6634875131444893, "lag_samples": -160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": -0.02, "neighbor_channel_1based": 18, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 26.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 34.23639754745607, "spectral_bandwidth_hz": 133.0259219598964, "arrival_rms_ac":

3.789669887549508e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 1.1081942262443245, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 15.547 dB. RMS AC: 6.2414e-06; MAD: 6.572e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [3.9406859227846705e-11, 3.686919725689976e-11, 3.7886381016766734e-11, 3.3753221210332634e-11], "max_min_power_ratio": 1.1674992138463918, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.002249274629244735, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 57.0, "prominence_db": 13.904407167870925, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch17: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste intermedio o sensible a los bloques de referencia", "Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [8.758010802845796, 10.858011789962516], "net_power_resolved": [true, true], "reference_power_ratio": 1.5487661785339077, "reference_stable": true, "normalized_block_powers": [0.00010871508902501126, 0.0001106308609993016, 0.00014777245332156725, 0.00016837425297824027], "normalization_amplitude": 0.00084027939010411 5, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [12.436155664111755, 23.60088732455999, 24.76942089610609, 20.858281106079314], "fraction": 1.0, "state": "ESTIMATED",

"persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.7316650352369634, "neighbor_consensus_rho": 0.4404618181175133, "pairs": [{"abs_rho": 0.7868267374044005, "signed_rho": 0.7868267374044005, "lag_samples": 68, "overlap_samples": 3932, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.0085, "neighbor_channel_1based": 16, "distance_m": 5.0}, {"abs_rho": 0.7966044796841409, "signed_rho": 0.7966044796841409, "lag_samples": -90, "overlap_samples": 3910, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.01125, "neighbor_channel_1based": 18, "distance_m": 5.0}, {"abs_rho": 0.5909644572500682, "signed_rho": 0.5909644572500682, "lag_samples": 141, "overlap_samples": 3859, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.017625000000000002, "neighbor_channel_1based": 15, "distance_m": 10.0}, {"abs_rho": 0.6765033330695264, "signed_rho": 0.6765033330695264, "lag_samples": -160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": -0.02, "neighbor_channel_1based": 19, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 24.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 34.82980017510507,

"spectral_bandwidth_hz": 146.89839045875775, "arrival_rms_ac": 3.181247349598518e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.9805123216236975, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 9.7878 dB. RMS AC: 9.8067e-06; MAD: 1.0206e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [7.676040348789043e-11, 7.811307155868131e-11, 1.0433761534932235e-10, 1.1888391677266095e-10], "max_min_power_ratio": 1.5487661785339084, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.0007723686141387321, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 48.638322338133854, "prominence_db": 29.332431982156177, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch18: NO_EVALUABLE

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste intermedio o sensible a los bloques de referencia"], "strict_grade": "NO_EVALUABLE", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [11.985711913320172, 13.118579413037663], "net_power_resolved": [true, true], "reference_power_ratio": 1.280292496981176, "reference_stable": true, "normalized_block_powers": [1.3531954548431004e-06, 1.3794489838262158e-06, 1.2207042244463954e-06, 1.0774483073819793e-06], "normalization_amplitude": 0.005880952347069979, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.7300459964143151, "neighbor_consensus_rho": 0.483379012943033, "pairs": [{"abs_rho": 0.7966044796841409, "signed_rho": 0.7966044796841409, "lag_samples": 90,

"overlap_samples": 3910, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.01125, "neighbor_channel_1based": 17, "distance_m": 5.0}, {"abs_rho": 0.7992634663517456, "signed_rho": 0.7992634663517456, "lag_samples": -54, "overlap_samples": 3946, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.00675, "neighbor_channel_1based": 19, "distance_m": 5.0}, {"abs_rho": 0.6634875131444893, "signed_rho": 0.6634875131444893, "lag_samples": 160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": 0.02, "neighbor_channel_1based": 16, "distance_m": 10.0}, {"abs_rho": 0.5010108249080505, "signed_rho": 0.5010108249080505, "lag_samples": -93, "overlap_samples": 3907, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.011625, "neighbor_channel_1based": 20, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 24.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 38.00300200206955, "spectral_bandwidth_hz": 172.6330988358601, "arrival_rms_ac": 2.8308361429666886e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 1.1318491998430784,

"quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 12.134 dB. RMS AC: 6.7969e-06; MAD: 6.9018e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [4.6801077411132156e-11, 4.7709071476480715e-11, 4.2218788645751026e-11, 3.726419672768119e-11], "max_min_power_ratio": 1.280292496981176, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.006732307331935537, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: []

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch19: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste condicional bajo", "Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [-1.4971175474821374, -0.5280592610212984], "net_power_resolved": [true, true], "reference_power_ratio": 1.1036606075211735, "reference_stable": true, "normalized_block_powers": [3.3962306315056037e-06, 3.319104588850188e-06, 3.4727218863740042e-06, 3.146548733105327e-06], "normalization_amplitude": 0.007475726772099733, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [18.774653805459124, 18.86071491260705, 26.32356804908641, 25.2693687713839], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause":

"UNIDENTIFIED"}, {"nominal_hz": 60.0, "block_prominence_db": [25.265161082215897, 21.461054480144778, 29.88827852262549, 25.09375061177533], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.6780035035366029, "neighbor_consensus_rho": 0.39563275886229504, "pairs": [{"abs_rho": 0.7992634663517456, "signed_rho": 0.7992634663517456, "lag_samples": 54, "overlap_samples": 3946, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.00675, "neighbor_channel_1based": 18, "distance_m": 5.0}, {"abs_rho": 0.6795036740036794, "signed_rho": 0.6795036740036794, "lag_samples": -53, "overlap_samples": 3947, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.006625, "neighbor_channel_1based": 20, "distance_m": 5.0}, {"abs_rho": 0.6765033330695264, "signed_rho": 0.6765033330695264, "lag_samples": 160, "overlap_samples": 3840, "at_search_boundary": true, "max_lag_s": 0.02, "lag_s": 0.02, "neighbor_channel_1based": 17, "distance_m": 10.0}, {"abs_rho": 0.55741129403888, "signed_rho": 0.55741129403888, "lag_samples": -67, "overlap_samples": 3933, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.008375, "neighbor_channel_1based": 21, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 22.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering

or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 61.75664532144822, "spectral_bandwidth_hz": 284.47847655209074, "arrival_rms_ac": 1.820896298794149e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.6973138853199511, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: -1.169 dB. RMS AC: 1.371e-05; MAD: 1.7543e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [1.8980341184412705e-10, 1.8549310797305403e-10, 1.9407823965339812e-10, 1.7584956673347138e-10], "max_min_power_ratio": 1.1036606075211732, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.009521669629509627, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 53.0, "prominence_db": 15.100430516831418, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 58.0, "prominence_db": 19.793244241958533, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch20: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste condicional bajo", "Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [1.7473689073441245, 4.125297037043314], "net_power_resolved": [true, true], "reference_power_ratio": 1.4368490074387055, "reference_stable": true, "normalized_block_powers": [0.0002079938425044039, 0.00016487128408393699, 0.00016971068892648698, 0.00014475692395484827], "normalization_amplitude": 0.0007733727688901126, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [10.91460796248458, 12.490397087830141, 18.370233784098733, 18.523961361776593], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause":

"UNIDENTIFIED"}, {"nominal_hz": 60.0, "block_prominence_db": [12.48889467830653, 11.424379585902667, 18.370233784098733, 18.523961361776593], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.7378928048221911, "neighbor_consensus_rho": 0.5441408675926676, "pairs": [{"abs_rho": 0.6795036740036794, "signed_rho": 0.6795036740036794, "lag_samples": 53, "overlap_samples": 3947, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.006625, "neighbor_channel_1based": 19, "distance_m": 5.0}, {"abs_rho": 0.8265321127456501, "signed_rho": 0.8265321127456501, "lag_samples": -8, "overlap_samples": 3992, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.001, "neighbor_channel_1based": 21, "distance_m": 5.0}, {"abs_rho": 0.5010108249080505, "signed_rho": 0.5010108249080505, "lag_samples": 93, "overlap_samples": 3907, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.011625, "neighbor_channel_1based": 18, "distance_m": 10.0}, {"abs_rho": 0.7962819356407027, "signed_rho": 0.7962819356407027, "lag_samples": -47, "overlap_samples": 3953, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.005875, "neighbor_channel_1based": 22, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 28.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No

filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.",

"spectral_centroid_hz": 83.81367364473353, "spectral_bandwidth_hz": 328.995465701059, "arrival_rms_ac": 1.761887046210093e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.8364721203267614, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 3.0271 dB. RMS AC: 1.0159e-05; MAD: 1.0858e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [1.244022486178273e-10, 9.861041185445711e-11, 1.0150488621550688e-10, 8.657990364595365e-11], "max_min_power_ratio": 1.436849007438706, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.014892526812826181, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 53.0, "prominence_db": 12.963474806031455, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 58.0, "prominence_db": 12.420029579963206, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch21: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste condicional bajo", "Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [null, null], "net_power_resolved": [false, false], "reference_power_ratio": 1.086764082889202, "reference_stable": true, "normalized_block_powers": [0.0005760201418504154, 0.0005556895300501579, 0.0006039034224960915, 0.0005780830836654718], "normalization_amplitude": 0.0012134758289903402, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [24.985587968164438, 31.228904534384565, 29.04557637183183, 25.70098509223825], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}, {"nominal_hz":

60.0, "block_prominence_db": [25.550606072983733, 31.228904534384565, 30.952691846068557, 25.70098509223825], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.8293235289178722, "neighbor_consensus_rho": 0.7300605239544454, "pairs": [{"abs_rho": 0.8265321127456501, "signed_rho": 0.8265321127456501, "lag_samples": 8, "overlap_samples": 3992, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.001, "neighbor_channel_1based": 20, "distance_m": 5.0}, {"abs_rho": 0.8321149450900944, "signed_rho": 0.8321149450900944, "lag_samples": -42, "overlap_samples": 3958, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.00525, "neighbor_channel_1based": 22, "distance_m": 5.0}, {"abs_rho": 0.55741129403888, "signed_rho": 0.55741129403888, "lag_samples": 67, "overlap_samples": 3933, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.008375, "neighbor_channel_1based": 19, "distance_m": 10.0}, {"abs_rho": 0.8576490814921628, "signed_rho": 0.8576490814921628, "lag_samples": -85, "overlap_samples": 3915, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.010625, "neighbor_channel_1based": 23, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 54.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival

accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 75.50773590012108, "spectral_bandwidth_hz": 259.7404497329653, "arrival_rms_ac": 2.391765434681978e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 1.3692779144974512, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: Sin valor dB. RMS AC: 2.9189e-05; MAD: 3.9319e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [8.482032457750581e-10, 8.182659403499827e-10, 8.892620342239199e-10, 8.51240976257459e-10], "max_min_power_ratio": 1.0867640828892025, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.00883127976584185, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 53.0, "prominence_db": 15.89840706539336, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 58.0, "prominence_db": 15.608207259507736, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

ch22: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste condicional bajo", "Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [null, null], "net_power_resolved": [false, false], "reference_power_ratio": 1.1661485854995646, "reference_stable": true, "normalized_block_powers": [0.00017449277924877317, 0.0002034845077008446, 0.00020196442256789214, 0.00020161312194309851], "normalization_amplitude": 0.001304750912822783, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [20.803120274264963, 24.052357174968844, 29.814992025639455, 29.538946157234342], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"},

{"nominal_hz": 60.0, "block_prominence_db": [24.8547955433599, 24.06002829759332, 31.0860835683512, 31.995452437225488], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause": "UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.8141984403653986, "neighbor_consensus_rho": 0.7413439774047135, "pairs": [{"abs_rho": 0.8321149450900944, "signed_rho": 0.8321149450900944, "lag_samples": 42, "overlap_samples": 3958, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.00525, "neighbor_channel_1based": 21, "distance_m": 5.0}, {"abs_rho": 0.8741615640680235, "signed_rho": 0.8741615640680235, "lag_samples": -42, "overlap_samples": 3958, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.00525, "neighbor_channel_1based": 23, "distance_m": 5.0}, {"abs_rho": 0.7962819356407027, "signed_rho": 0.7962819356407027, "lag_samples": 47, "overlap_samples": 3953, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.005875, "neighbor_channel_1based": 20, "distance_m": 10.0}, {"abs_rho": 0.6824535480780894, "signed_rho": 0.6824535480780894, "lag_samples": 0, "overlap_samples": 4000, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.0, "neighbor_channel_1based": 24, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 54.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of

first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 118.96110928491753, "spectral_bandwidth_hz": 425.0081749757613, "arrival_rms_ac": 1.5546541288501512e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": 0.8900344207875207, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: Sin valor dB. RMS AC: 1.8286e-05; MAD: 2.496e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [2.9705213539135494e-10, 3.464069275062536e-10, 3.438191726623904e-10, 3.4322112728075023e-10], "max_min_power_ratio": 1.1661485854995641, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.007488486076692456, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 53.0, "prominence_db": 14.412046367388173, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}, {"state": "CANDIDATE", "nominal_hz": 60.0, "fundamental_hz": 60.0, "harmonic": 1, "estimated_hz": 58.0, "prominence_db": 13.775186973637846, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch23: REGULAR

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: Líneas espectrales candidatas próximas a red/armónicos; comprobar persistencia; snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste condicional bajo", "Línea candidata persistente entre bloques; revisar su efecto, sin atribuir causa eléctrica"], "strict_grade": "REGULAR", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [4.071236237223015, 4.812803879851366], "net_power_resolved": [true, true], "reference_power_ratio": 1.1337976279763506, "reference_stable": true, "normalized_block_powers": [6.862209649763994e-06, 7.004583905124206e-06, 7.780357023578841e-06, 7.5610825064979234e-06], "normalization_amplitude": 0.0032932120375335217, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [{"nominal_hz": 50.0, "block_prominence_db": [13.553505234849926, 21.15699139627585, 22.397835592968498, 19.45984135216904], "fraction": 1.0, "state": "ESTIMATED", "persistent_candidate": true, "cause":

"UNIDENTIFIED"}], "estimated_similarity": {"state": "ESTIMATED", "grade": "REGULAR", "median_abs_rho": 0.8576490814921628, "neighbor_consensus_rho": 0.6973641956883843, "pairs": [{"abs_rho": 0.8741615640680235, "signed_rho": 0.8741615640680235, "lag_samples": 42, "overlap_samples": 3958, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.00525, "neighbor_channel_1based": 22, "distance_m": 5.0}, {"abs_rho": 0.7020705809042155, "signed_rho": 0.7020705809042155, "lag_samples": -96, "overlap_samples": 3904, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": -0.012, "neighbor_channel_1based": 24, "distance_m": 5.0}, {"abs_rho": 0.8576490814921628, "signed_rho": 0.8576490814921628, "lag_samples": 85, "overlap_samples": 3915, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.010625, "neighbor_channel_1based": 21, "distance_m": 10.0}], "reasons": ["Vecinos sin consenso: cambio lateral o referencia heterogénea; revisar"], "spectral_median_hz": 54.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 96.40852430974374, "spectral_bandwidth_hz": 346.7277080128886, "arrival_rms_ac": 1.731582627635917e-05, "low_correlation_fraction": 0.0, "consensus_channels_1based": [], "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"],

"local_amplitude_ratio": 1.113805698323797, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 4.3262 dB. RMS AC: 8.9926e-06; MAD: 1.0243e-05 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [7.442234848992094e-11, 7.596643224562143e-11, 8.437988218630171e-11, 8.200179621136844e-11], "max_min_power_ratio": 1.1337976279763506, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.007155094981932408, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: [{"state": "CANDIDATE", "nominal_hz": 50.0, "fundamental_hz": 50.0, "harmonic": 1, "estimated_hz": 53.0, "prominence_db": 13.720180647956884, "bin_width_hz": 2.0, "frequency_uncertainty_hz": null, "cause": "UNIDENTIFIED"}]

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

Acciones: Comprobar ruido en toma independiente antes de seleccionar un notch; Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

ch24: NO_EVALUABLE

Integridad 100 %. SNR válida: MISSING / Sin valor dB. Clipping: MISSING / Sin valor (fracción).

Motivos: snr: Falta ventana independiente de ruido o de señal; clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping; Comparabilidad de componentes/receptores no declarada

Estimaciones y soporte diagnóstico: {"grade": "REGULAR", "color": "#b77900", "policy_id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "scope": "CONDITIONAL_SAMPLE_SCREENING_NOT_FIELD_ACCEPTANCE", "coverage_fraction": 0.75, "domains": {"integrity": "SUPPORTED", "morphology": "SUPPORTED", "contrast": "ESTIMATED", "similarity": "MISSING"}, "minimum_coverage": 0.75, "reasons": ["Contraste condicional bajo"], "strict_grade": "NO_EVALUABLE", "missing_metadata": ["snr: Falta ventana independiente de ruido o de señal", "clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping", "Comparabilidad de componentes/receptores no declarada"], "estimated_contrast": {"state": "ESTIMATED", "db_bounds": [3.0228585833842287, 4.404610858658914], "net_power_resolved": [true, true], "reference_power_ratio": 1.249971588945676, "reference_stable": true, "normalized_block_powers": [1.5899986099113587e-06, 1.3665636021912305e-06, 1.2760720384542394e-06, 1.5950537935157933e-06], "normalization_amplitude": 0.0055780839174985886, "reference_kind": "POST_SIGNAL", "noise_independence_verified": false, "interval_kind": "REFERENCE_BLOCK_SENSITIVITY_NOT_CI", "reason": "Contraste condicional usando todos los bloques; no valida transferencia del ruido ni SNR del arribo"}, "morphology": {"state": "MEASURED", "candidate": false, "longest_run": 1, "plateau_fraction": 0.0, "adc_rails_estimated": false, "reason": "Sin mesetas extremas detectadas; esto no demuestra ausencia de clipping físico"}, "line_persistence": [], "estimated_similarity": {"state": "MISSING", "grade": "REGULAR", "median_abs_rho": null, "neighbor_consensus_rho": null, "pairs": [{"abs_rho": 0.7020705809042155, "signed_rho": 0.7020705809042155, "lag_samples": 96, "overlap_samples": 3904, "at_search_boundary": false, "max_lag_s": 0.02,

"lag_s": 0.012, "neighbor_channel_1based": 23, "distance_m": 5.0}, {"abs_rho": 0.6824535480780894, "signed_rho": 0.6824535480780894, "lag_samples": 0, "overlap_samples": 4000, "at_search_boundary": false, "max_lag_s": 0.02, "lag_s": 0.0, "neighbor_channel_1based": 22, "distance_m": 10.0}], "reasons": ["Vecinos comparables insuficientes; sin decisión local"], "spectral_median_hz": 30.0, "baseline_to_arrival_rms": null, "baseline_relative_to_neighbors": null, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "spectral_centroid_hz": 137.5842777047187, "spectral_bandwidth_hz": 475.0856548622073, "arrival_rms_ac": 1.2213838148928332e-05, "actions": ["Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo"], "local_amplitude_ratio": null, "quality_applicable": false, "instrumental_comparability_verified": false, "scope": "LOCAL_SHAPE_SCENARIO_NOT_INSTRUMENT_DIAGNOSIS"}, "eligible_for_acceptance": false}

Ruido: CONDITIONAL. SNR condicional: 3.4957 dB. RMS AC: 6.7891e-06; MAD: 7.2249e-06 V.

Estacionariedad interna: {"state": "COMPATIBLE_WITHIN_POST_SIGNAL_REFERENCE", "block_powers": [4.947283885050734e-11, 4.252065407399623e-11, 3.970500724123551e-11, 4.963013099042678e-11], "max_min_power_ratio": 1.2499715889456775, "transfer_to_signal_verified": false}

Baja frecuencia: {"state": "PARTIAL", "requested_band_hz": [1.0, 15.0], "supported_band_hz": [6.0, 15.0], "noise_power_to_signal_total_power": 0.03311213030833391, "bin_width_hz": 2.0, "cause": "UNIDENTIFIED", "minimum_cycles": 3.0}

Líneas candidatas: []

Similitud: MISSING; |rho|=Sin valor; consenso=Sin valor

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Acciones: Comparar ventanas/ramas de llegada y réplicas; verificar orden, coordenadas, ganancia y acople en campo

Coherencia LF entre canales contiguos

ch01 / ch02: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch02 / ch03: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch03 / ch04: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch04 / ch05: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch05 / ch06: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch06 / ch07: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch07 / ch08: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch08 / ch09: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch09 / ch10: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch10 / ch11: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch11 / ch12: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch12 / ch13: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch13 / ch14: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch14 / ch15: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch15 / ch16: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch16 / ch17: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch17 / ch18: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch18 / ch19: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch19 / ch20: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

ch20 / ch21: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch21 / ch22: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch22 / ch23: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

ch23 / ch24: {"state": "MISSING", "value": null, "required_duration_s": 7.5, "reason": "Se requieren cuatro segmentos y tres ciclos por segmento; no rellenar con ceros"}

Cobertura metodológica y evidencia pendiente

Etapa	Estado
integrity	EXECUTED
snr_clipping	EXECUTED_WITH_APPLICABILITY_GUARDS
noise	EXECUTED_CONDITIONAL
similarity	EXECUTED_WITH_APPLICABILITY_GUARDS
filtering	NOT_APPLIED_RAW_PRESERVED
repeatability	NOT_REQUESTED_REQUIRES_SOURCE_GROUPS
reciprocity	NOT_REQUESTED_REQUIRES_PHYSICAL_PAIRS

Comparabilidad de componentes/receptores no declarada

clip: Rails/escala desconocidos; sospecha morfológica no confirma clipping

snr: Falta ventana independiente de ruido o de señal

Primeros arribos y ramas TRS

Receptor	Offset (m)	Llegada (s)	Sigma (s)	Estado
ch01	15	0.010375	Sin valor	UNCALIBRATED
ch02	20	0.00325	Sin valor	UNCALIBRATED
ch03	25	0.003125	Sin valor	UNCALIBRATED
ch04	30	0.016875	Sin valor	UNCALIBRATED
ch05	35	0.02425	Sin valor	UNCALIBRATED
ch06	40	Sin valor	Sin valor	AMBIGUOUS
ch07	45	Sin valor	Sin valor	AMBIGUOUS
ch08	50	Sin valor	Sin valor	AMBIGUOUS
ch09	55	Sin valor	Sin valor	AMBIGUOUS
ch10	60	Sin valor	Sin valor	AMBIGUOUS
ch11	65	Sin valor	Sin valor	AMBIGUOUS

GeoQC - TRS | Calidad por traza y tendido

DRAFT - DIAGNOSTIC ONLY - NOT_DEMONSTRATED - eligible_for_acceptance=false

Digest: 96ce4c9eea26f54078c5ad51b5f4441fa5ee1a5b66f470ba99539095c0654525

Receptor	Offset (m)	Llegada (s)	Sigma (s)	Estado
ch12	70	Sin valor	Sin valor	AMBIGUOUS
ch13	75	Sin valor	Sin valor	AMBIGUOUS
ch14	80	Sin valor	Sin valor	AMBIGUOUS
ch15	85	Sin valor	Sin valor	AMBIGUOUS
ch16	90	Sin valor	Sin valor	AMBIGUOUS
ch17	95	Sin valor	Sin valor	AMBIGUOUS
ch18	100	Sin valor	Sin valor	AMBIGUOUS
ch19	105	Sin valor	Sin valor	AMBIGUOUS
ch20	110	Sin valor	Sin valor	AMBIGUOUS
ch21	115	Sin valor	Sin valor	MISSING
ch22	120	Sin valor	Sin valor	MISSING
ch23	125	Sin valor	Sin valor	AMBIGUOUS
ch24	130	Sin valor	Sin valor	AMBIGUOUS

Rama LEFT: {"side": "LEFT", "state": "MISSING", "reason": "Menos de cinco offsets únicos con onset por lado"}

Rama RIGHT: {"side": "RIGHT", "receiver_ids": ["ch01", "ch02", "ch03", "ch04", "ch05"], "fit": {"state": "MODEL_INAPPLICABLE", "reason": "BIC gaussiano no válido para pérdidas robustas"}, "parameters": {"state": "MODEL_INAPPLICABLE", "reason": "Modelo horizontal dos capas no verificado"}, "inversion": {"state": "MODEL_INAPPLICABLE", "reason": "Inversión requiere modelo/ROI y sigmas temporales calibradas"}}

Procedencia y protocolo congelado

{"config": null, "noise": {"block_power_ratio_warn": 4.0, "harmonics": 5, "line_prominence_db": 10.0, "low_band_hz": [1.0, 15.0], "low_ratio_warn": 0.5, "mains_candidates_hz": [50.0, 60.0], "minimum_cycles": 3.0, "noise": {"end_s": 2.0, "schema_version": "1.4", "start_s": 1.5}, "noise_blocks": 4, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy.", "reference_kind": "POST_SIGNAL", "schema_version": "1.4", "signal": {"end_s": 0.5, "schema_version": "1.4", "start_s": 0.0}}, "operational_policy": {"contrast_good_db": 12.0, "contrast_review_db": 6.0, "id": "TRS_SAMPLE_SCREENING_V1_DRAFT", "line_persistence_min_fraction": 0.75, "line_prominence_db": 10.0, "max_reference_power_ratio": 4.0, "minimum_coverage": 0.75, "plateau_min_run": 3, "reference": "docs/trs_operational_quality.md; fixed declared windows; local screening thresholds, not field-calibrated acceptance. Metadata and strict QC preserved.", "reference_blocks": 4, "schema_version": "1.4"}, "protocol": {"amplitude_model_ref": null, "amplitude_ratio_bounds": [0.4, 2.5], "analysis": {"end_s": 2.0, "schema_version": "1.4", "start_s": 0.0}, "antialias_verified": null, "expected_relative_amplitude": null, "frequency_band_hz": [40.0, 160.0], "group_velocity_ref": null, "group_velocity_verified": false, "guard_s": 0.0, "line_power_ratio": 20.0, "log_mad_limit": 3.5, "mains_hz": null, "mains_search_hz": 1.0, "min_group_velocity_ms": null, "noise": {"end_s": 2.0, "schema_version": "1.4", "start_s": 1.5}, "noise_independent": false,

"noise_stationarity": "UNKNOWN", "nyquist_margin": 0.8, "post_margin_s": 0.0, "provenance": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy.", "quality_policy": {"clip_bad_min_fraction": 0.01, "clip_good_max_fraction": 0.001, "id": "TRACE_QUALITY_V1_DRAFT", "schema_version": "1.4", "snr_bad_below_db": 6.0, "snr_good_min_db": 12.0}, "rail_scale_to_data": null, "rail_tolerance": 0.0, "schema_version": "1.4", "signal": {"end_s": 0.5, "schema_version": "1.4", "start_s": 0.0}, "similarity": {"arrival": {"end_s": 0.5, "schema_version": "1.4", "start_s": 0.0}, "bad_below_rho": 0.5, "bad_neighbor_fraction_min": 0.75, "baseline": null, "baseline_ratio_warn": 5.0, "comparable_channels": false, "max_lag_s": 0.02, "max_neighbors": 4, "min_apparent_velocity_ms": 100.0, "min_neighbors": 3, "neighbor_consensus_min": 0.8, "neighbor_radius_m": 10.1, "reference": "KM40 F1 EXPLORATORY: analysis [0,2) s; signal [0,0.5) s and posterior reference [1.5,2) s reuse the declared picking windows. Reference assumed signal-free; independence and transfer to signal NOT verified. No ADC rails/scale verified. No instrumental comparability verified. No filtering or calibration of first-arrival accuracy. Neighborhood/lag are draft scenarios; do not assert comparability.", "schema_version": "1.4", "warn_below_rho": 0.8}, "stationarity_ref": null, "wavelet_duration_s": 0.0, "welch_segment_samples": 256}, "schema_version": "1.4"}

Procedencia de la ejecución

{"acquisition_revision_id": "54eeb567-c3b8-4826-ad8c-fd6960b01b0a", "eligible_for_acceptance": false, "engine_reference": {"code_hash": "698c9a84c02ec0b3690aeb81f1731d30208cbd3594f4f52af51a1a6ba21bffab", "masw_estimator_hash": "046537818baeba8a24697d7a3634baf6c2b32f980c73c7169bd59fe63bf85292", "package_version": "0.0.3", "runtime": [{"name": "machine", "version": "AMD64"}, {"name": "numpy", "version": "1.26.4"}, {"name": "platform", "version": "win32"}, {"name": "pydantic", "version": "2.12.5"}, {"name": "python", "version": "3.12.3"}, {"name": "scipy", "version": "1.15.2"}], "trs_estimator_hash": "efdd70d2523497f7e1b71bf71a186a5c9a1829f237671dbc922842607aebe70f"}, "input_manifest": {"protocol_hash": "445403521e56a183e5613175a387e6abd51619c827d56233a4670ad322a79f4a", "shots": [{"headers_hash": "bc1dd69333c9afa94dfe0834e9ba1fcd1ff6b8bc34657efd63157cf1331dea8f", "input_hashes": ["99110dd27715ccb975a23c8ea6c86ca5c2386ba6b587e88c64ef8e82dc8d0130", "cdfd4ea6c885094e06726099086048d301cd34693e11f9ff625a19ef7a0059f5"], "scientific_shot_hash": "47ad7edbddaa4bb8ce67661287c89f6e324db7ebcc9b38efbf3f486960d7cf1d", "shot_id": "KM40_F1_TRS_EXPLORATORY_ASSUMPTIONS"}]}, "sufficiency": {"status": "NOT_DEMONSTRATED"}}

Las matrices, PSD y vectores completos se conservan en el JSON científico y en el HTML autónomo del mismo digest. Este PDF presenta las gráficas, métricas, motivos y criterios de calidad. La revisión humana se consulta por separado.